

PURPOSE

Sepsis is the leading cause of in-hospital mortality in the United States, responsible for 1 in 5 deaths globally (Singer et al., 2016). In the US alone, sepsis accounts for **\$62 billion** in annual healthcare expenditures, more than any other disease. Despite this burden, there are **no FDA-approved drugs** that specifically target the underlying molecular driver of sepsis: dysregulated protease cascades.

During the onset of sepsis, infection triggers an overwhelming immune response that activates dozens of protease enzymes simultaneously. These proteases normally serve critical regulatory functions in immunity, coagulation, and wound healing. In sepsis, however, their coordinated dysregulation drives a self-amplifying loop causing **irreversible multi-organ failure** (Gando et al., 2016).

Current solutions for Inhibitor Discovery Are Insufficient: Static predictions and Low Accuracy



Figure 3: Current Detection method and treatment of Sepsis (MDPI, 2019)



Figure 4: Length, Cost, and Inaccuracies of current Discovery. (Ze Clinics, 2025)

THE HUMAN IMPACT OF SEPSIS
1.7 million+
US deaths attributable to sepsis annually

Standardized mortality ratio (SMR) is a measure of the quality of care. A SMR of 1.0 indicates that the observed mortality rate is equal to the expected mortality rate. A SMR of 1.5 indicates that the observed mortality rate is 50% higher than the expected mortality rate.

Figure 1: Sepsis large-scale impact on United States residents in 2025 the Assembly, 2025)

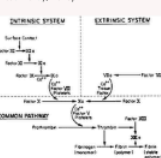
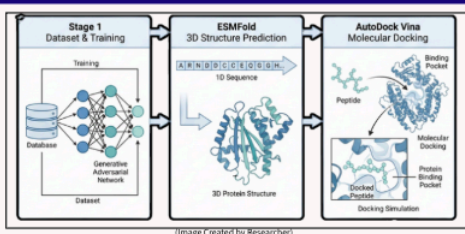


Figure 1: Sepsis large-scale impact on United States residents in 2025 the Assembly, 2025)

Figure 2: Coagulation Cascade (Science Direct, 2020)

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Solution Overview



(Image Created by Researcher)

Dataset

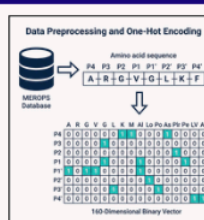


Figure 4: Step by step diagram of the Data Preprocessing and One-Hot Encoding prior to training of the pipeline (Image Created by Researcher using Canva).



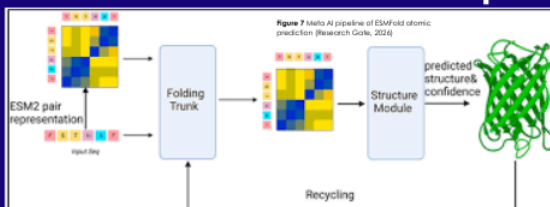
Figure 5: The RCSB Protein Data Bank

Data Collection

- 11,551 experimentally confirmed proteolytic cleavage events extracted from MEROPS database (Release 12.4)
- High-resolution 3D crystal structures for 27 target proteases sourced from the RCSB Protein Data Bank (PDB)
- Four Pathological Protease Classes**
 - Matrix Metalloproteinases (MMPs): Tissue-destructive
 - Caspases: Drivers of cellular apoptosis
 - Serine Proteases: Coagulation and inflammatory cascades
 - Cysteine Proteases: Cytotoxic pathways

SPECTRA: De Novo Broad-Spectrum Sepsis Inhibitor Design Across 27 Protease Targets via a Multi-Architecture Generative Adversarial Pipeline

Pipeline



Training + Architecture

A multi-branched ensemble GAN pipeline ran three distinct algorithms simultaneously to prevent mode collapse common in unimodal networks.

Simulation

Generated sequences are passed into Meta AI's ESMFold language model, which predicts full-atom 3D folded geometries in 10-15 seconds per structure.

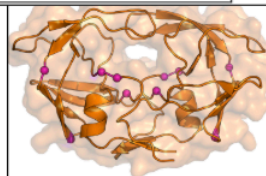


Figure 8: In Silico Structural Simulation. Vectorized representation of the peptide executing ESMFold atomic prediction on a 1D sequence. The model predicts the 3D folded geometry of the peptide, which is then used to simulate its interaction with the target protease. (Created by Researchers using ESMFold)

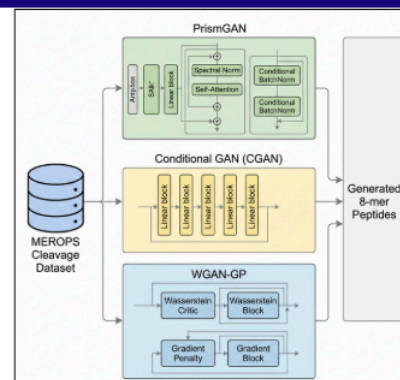


Figure 6: Ensemble GAN Architecture Topology. Depiction of the three parallel Deep Learning networks built to evaluate the MEROPS dataset. PrismGAN (top) incorporates Self-Attention and Spectral Normalization, Conditional GAN (middle) structures linear transformations, and WGAN-GP (bottom) penalizes gradient instability for robust generative output across all axes. (Created by Researcher using Canva)

Model Performance



Figure 9: Per-Protease Head-to-Head Algorithm vs. Nature Performance. Created By Researchers using Matplotlib)

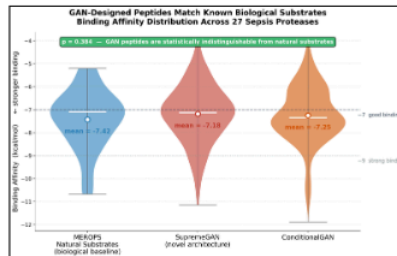


Figure 10: Binding Affinity Distribution & Equivalence (Comparison between baseline and GAN's). (Created By Researchers using Matplotlib)

Inhibitor Optimization & ADMET Profiling



Figure 11: Inhibitor Optimization Structure and Process (Created by researcher using Canva)

Highly-rated reversible peptide substrates were computationally upgraded into irreversible covalent inhibitors by attaching a chemical warhead designed to permanently block the enzyme's active site.



Figure 12: ADMET Profiling factors (Created by Reachers using Canva)

These optimized covalent inhibitors underwent comprehensive computational profiling to ensure highly favorable Absorption, Distribution, Metabolism, Excretion, and Toxicity limits for safe human administration during systemic sepsis.

Discussion

Performance Analysis

The procedurally AI-generated candidate panel culminated into 28 total verified "excellent binders". Panels binding affinity spans **-4.93** to a highly potent **-11.88 kcal/mol**.

Statistical Testing

A two One-Sided Test algorithm (TOST) was used with a severity margin of $\delta = \pm 0.75$ kcal/mol (Trott & Olson, 2010). The pipeline proved to be equivalent MEROPS baselines (bounds perfectly measured at statistically significant limits of $p_{lower} < 0.001$, $p_{upper} = 0.011$).

Future Development

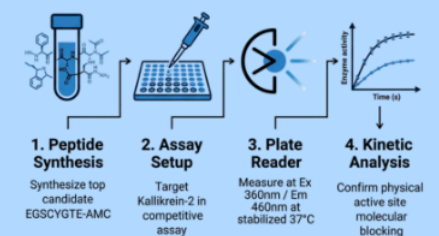


Figure 11: Image Created by Researcher using Canva

Empirical Wet-Lab Validation:

- 9.72 kcal/mol *EGSCYGT-AMC* top pipeline candidate.

Molecular Dynamics Trajectories

Molecular Dynamics Trajectory:

- Simulating peptide within fully explicit physical solvent simulations to detail precise MM-GBSA kinetic stability (Fig 12)

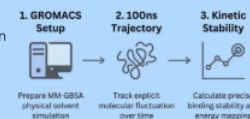


Figure 12: Image Created by Researcher using Canva

Conclusion



AI Outperforms Nature: Generative models exceeded mammalian evolution in 25 out of 27 targets.

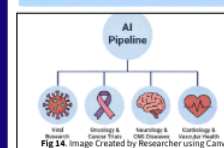


Multi-Target Efficacy: The single pipeline effectively neutralizes dozens of pathways at once.



Universal Architecture: Framework seamlessly redirects to other multi-pathway diseases.

Figure 13: Image Created by Researcher using Canva



The multi-architecture GAN framework is universally scalable, easily translating from Sepsis to combat Cancers and Alzheimer Pathology (Fig 14)